

Examples

Determine whether the following argument is valid:

$$\begin{array}{l} (p \wedge q) \vee r \quad \textcircled{1} \\ r \rightarrow s \quad \textcircled{2} \\ \hline \therefore p \vee s \end{array}$$

$$(\textcircled{1} \wedge \textcircled{2}) \rightarrow p \vee s$$

Resolution

$$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array}$$

$$\begin{array}{l} \textcircled{1} \quad r \vee (p \wedge q) \equiv (r \vee p) \wedge (r \vee q) \\ \textcircled{2} \quad \neg r \vee s \end{array}$$

$$(r \vee p) \wedge (r \vee q) \wedge (\neg r \vee s)$$

$$\begin{array}{l} (p \vee s) \wedge (r \vee q) \\ \hline (p \vee s) \end{array} \quad \downarrow \text{simplification}$$

Rules of inference: Summary

- Modus ponens: $p \rightarrow q$ and p therefore q
 - Modus tollens: $p \rightarrow q$ and $\neg q$ therefore $\neg p$
 - Hypothetical syllogism: $p \rightarrow q$ and $q \rightarrow r$ therefore $p \rightarrow r$
 - Disjunctive syllogism: $p \vee q$ and $\neg p$ therefore q
 - Addition: p therefore $p \vee q$
 - Simplification: $p \wedge q$ therefore q
 - Conjunction: p and q therefore $p \wedge q$ (trivially!)
 - Resolution: $p \vee q$ and $\neg p \vee r$ therefore $q \vee r$
- 

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$$\begin{array}{l} p \rightarrow r \quad (1) \\ q \rightarrow r \quad (2) \\ \neg r \quad (3) \\ \hline \therefore \underline{p \vee q} \end{array}$$

① and ③ $(p \rightarrow r) \wedge \neg r$ * Modus Tollens

$$\neg p \wedge (2)$$

$$\neg p \wedge (\neg q \vee r)$$

$$(\neg p \wedge \neg q) \vee (\neg p \wedge r) \quad \downarrow \text{distrib.}$$

$$\neg(\underline{p \vee q}) \vee (\neg p \wedge r)$$

FALSE

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- $$\frac{\exists x P(x)}{\therefore P(c) \text{ for some } c \text{ in the domain}}$$

Sec 1.8: Introduction to proofs

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 - ▶ Proposition: less important mathematical result
 - ▶ Lemma: a result that is needed to prove a theorem
 - ▶ Corollary: a result that directly follows from a theorem

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- Proof techniques:
 - ▶ Direct proof
 - ▶ Proof by contraposition
 - ▶ Proof by contradiction
 - ▶ Proof by cases

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P	Q	$P \rightarrow Q$
T	T	T
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Direct proof

$\overbrace{\quad}^p$ $\overbrace{\quad}^q$
Example. If n is an odd integer, then n^2 is odd.

Let n be an arbitrary odd integer.

$$\exists k \text{ s.t. } n = \underline{2k+1}$$

$$\begin{aligned} n^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= \underline{2(2k^2 + 2k)} + \underline{1} \\ &\quad \text{integer.} \end{aligned}$$

Since $2k^2 + 2k$ is an integer, n^2 has the form of $2m+1$ ($m=2k^2+2k$) $\Rightarrow n^2$ is odd

QED

Direct proof

Example. If $\overbrace{0 \leq x \leq 2}^P$, then $\overbrace{-x^3 + 4x + 1}_{Q} > 0$.

Assume $0 \leq x \leq 2$

$$-x^3 + 4x + 1 = x(4 - x^2) + 1$$

Since $x \leq 2$

$$x^2 \leq 4$$

$$0 \leq 4 - x^2$$

$$0 \leq x(4 - x^2)$$

$$0 < 1 \leq x(4 - x^2) + 1$$



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- We can prove $P(c) \rightarrow Q(c)$ by proving $\neg Q(c) \rightarrow \neg P(c)$
- **Proof strategy:** To prove $P(c) \rightarrow Q(c)$, we need to show that if conclusion $Q(c)$ is F, then $P(c)$ is false too